

AI Voice Agent Unit Economics & Cost Optimisation Plan

Mahavitaran (MSEDCL) customer-care voice agent · Marathi / Hindi / English · Exotel telephony · Sarvam speech stack · Gemini reasoning · AWS Mumbai

₹2.61 variable cost per live call-minute	₹2.95 all-in per minute @ 10k min/month	₹1.95 optimised per minute (no quality trade-off)	up to 49% cost reduction identified
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Executive summary

Every minute a customer talks to the voice agent consumes four metered services: Exotel carries the call, Sarvam converts speech to text and text back to speech, and Google Gemini decides what to say. A fixed AWS server hosts the pipeline. At today's rates the variable cost is ₹2.61 per live minute; with server cost amortised this is ₹2.95/min at 10,000 minutes per month, falling to ₹2.64/min at 100,000 minutes as fixed cost dilutes.

Text-to-speech is the single largest cost driver (44% of the variable cost), followed by the language model (40%). Both have concrete, low-risk optimisation levers already identified — prompt caching, response-length discipline (already enforced in the agent), and vendor tier choices — that bring the per-minute cost down by 25% with no quality impact, and by up to 49% if A/B tests approve two vendor downgrades.

Where each rupee goes (variable cost per live minute)



■ TTS (Sarvam Bulbul v3) ₹1.15 ■ LLM (Gemini 2.5 Flash) ₹1.05 ■ STT (Sarvam Saarika) ₹0.21 ■ Telephony (Exotel) ₹0.20

All amounts in INR. USD converted at ₹88/\$ (stated assumption). Rates verified against published vendor price lists and the company's own Exotel rate card, 13 Jul 2026. Usage profile measured from the production pipeline (see page 2).

1 · Vendor rate card (verified)

Service	Vendor / SKU	List rate	Billing unit
Telephony (incoming)	Exotel — company rate card	₹0.20 / min	per pulse
Telephony (outgoing)	Exotel — company rate card	₹0.60 / min	per pulse
Speech-to-text	Sarvam Saarika v2.5	₹30 / audio-hour	per second of audio
Text-to-speech	Sarvam Bulbul v3	₹30 / 10,000 chars	per character
Text-to-speech (alt.)	Sarvam Bulbul v2	₹15 / 10,000 chars	per character
LLM — input	Gemini 2.5 Flash	\$0.30 / 1M tokens (₹26.4)	per token
LLM — output	Gemini 2.5 Flash	\$2.50 / 1M tokens (₹220)	per token
LLM — cached input	Gemini 2.5 Flash (implicit cache)	\$0.075 / 1M tokens (₹6.6)	per token
Hosting	AWS EC2 t3.medium, Mumbai	≈ ₹3,400 / month	fixed (incl. EBS + egress)

2 · Usage profile of one live call-minute

Measured from the production pipeline's behaviour (voice-activity-gated STT, per-sentence TTS with caching, one LLM round per conversational turn plus tool lookups):

Driver	Consumption per live minute	Why
STT audio	≈ 25 seconds	caller speaks ~45% of the call; VAD sends only voiced audio
TTS characters	≈ 380 billed chars	agent speaks ~28 s; short-reply policy; ~10% cache hits
LLM input tokens	≈ 35,000	4 turns × (system prompt + tools + history) + tool rounds
LLM output tokens	≈ 500	replies are deliberately 1–3 short sentences
Telephony	1 minute	incoming leg only (no outbound dialling in this flow)

3 · Cost per live call-minute

Component	Calculation	₹ / minute	Share
Text-to-speech	380 chars × ₹30/10k	₹1.15	44%
LLM (Gemini 2.5 Flash)	35k in + 500 out tokens	₹1.05	40%
Speech-to-text	25 s × ₹30/hr	₹0.21	8%
Telephony (incoming)	1 min × ₹0.20	₹0.20	8%
Variable cost / minute		₹2.61	100%

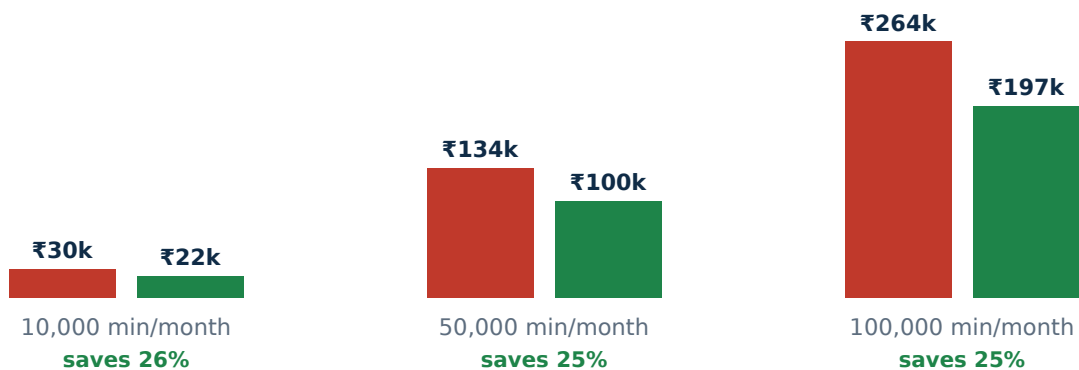
Server cost is fixed and dilutes with scale: add ₹3,400/month ÷ monthly minutes (₹0.34/min at 10k → ₹0.03/min at 100k).

4 · Monthly cost at three volumes

	Pilot — 10,000 min	Scale — 50,000 min	Rollout — 100,000 min
Sarvam TTS	₹11,500	₹57,500	₹115,000
Gemini LLM	₹10,500	₹52,500	₹105,000
Sarvam STT	₹2,100	₹10,500	₹21,000
Exotel telephony	₹2,000	₹10,000	₹20,000
AWS hosting (fixed)	₹3,400	₹3,400	₹3,400
Total / month	₹29,500	₹133,900	₹264,400
All-in ₹ per minute	₹2.95	₹2.68	₹2.64

Current vs optimised monthly spend

Optimised = conservative programme only (no voice/model downgrades) — see page 4.



Sensitivity notes for finance

Pulse billing: Exotel bills per pulse — a 61-second call can bill as 2 minutes on a 60-second pulse. Confirming per-second (or 30-second) pulse on the rate card is worth 5–15% of telephony spend.

Call-length discipline is already engineered: the agent speaks in 1–3 sentence turns, disconnects after three unanswered silence prompts, and resolves most queries in 2–4 minutes — the largest uncontrolled cost variable is average call duration, which the product already actively minimises.

FX exposure: only the Gemini line (~40% of variable cost) is dollar-denominated. A ₹1 move against the dollar shifts the per-minute cost by roughly ₹0.012 (~0.5%).

Outbound calls cost 3× the incoming rate (₹0.60 vs ₹0.20) — keep callback features on missed-call / incoming flows wherever possible.

5 · Cost-optimisation programme

Phase 1 measures are engineering work only — no user-visible change, no vendor migration.

Phase 2 measures need a short quality A/B before adoption.

#	Measure	Mechanism	Saves (₹/min)	Risk
P1	Gemini prompt caching	The ~4.7k-token system prompt repeats every turn; Gemini's implicit cache re-bills repeated prefixes at 25% of list price	0.35–0.45	None
P1	Trim context window	Cap history at ~24 messages and slim tool schemas (–20% input tokens)	0.10–0.15	None
P1	Wider TTS caching	Pre-warm & cache recurring lines (greetings, confirmations, closings) — cache hits already cost ₹0	0.08–0.12	None
P1	AWS savings plan	1-yr no-upfront compute savings plan (–28%); optionally Graviton t4g.medium (–20% more)	₹950+/mo fixed	None
P1	Exotel pulse audit	Confirm per-second pulse; negotiate volume slab at 50k+ min	0.02–0.05	None
P2	Gemini 2.5 Flash-Lite	\$0.10/\$0.40 per M tokens (vs \$0.30/\$2.50) for this narrow, tool-driven domain	up to 0.55	A/B quality gate
P2	Bulbul v2 voice	₹15/10k chars — half the v3 rate	up to 0.55	A/B voice gate

Scenario	Variable ₹/min	vs today	Monthly @ 50k min
Today	₹2.61	—	₹133,900
Phase 1 only (zero risk)	₹1.95	–25%	₹99,950
Phase 1 + 2 (post A/B)	₹1.32	–49%	₹68,450

6 · Assumptions & verification

- All figures are exclusive of GST. AWS India, Sarvam and Exotel invoices add 18% GST; Gemini (billed from Google's US entity) attracts 18% IGST under reverse charge on imported services. For a GST-registered company the full 18% is recoverable as input tax credit, so GST is a working-capital item, not a cost — the per-minute economics in this document are the true P&L cost.

- USD/INR at ₹88; Sarvam and Exotel bill in INR, Gemini in USD.

- Sarvam list rates from sarvam.ai/api-pricing; Gemini rates from ai.google.dev/gemini-api/docs/pricing; AWS t3.medium on-demand Mumbai region; Exotel rates from the company's own rate card (₹0.20 in / ₹0.60 out).

- Usage profile (25 s STT audio, 380 TTS chars, 35k/500 LLM tokens per live minute) is derived from the production pipeline design; the server now logs per-stage consumption, so these can be replaced by measured 30-day averages before contract negotiations.

- One t3.medium comfortably serves the pilot and scale volumes; at sustained 100k+ min/month, budget a second instance (+₹3,400/month, already immaterial at that scale).

- Prices checked 13 Jul 2026; AI vendor pricing changes frequently — re-verify quarterly.